

3D Printing and Modeling

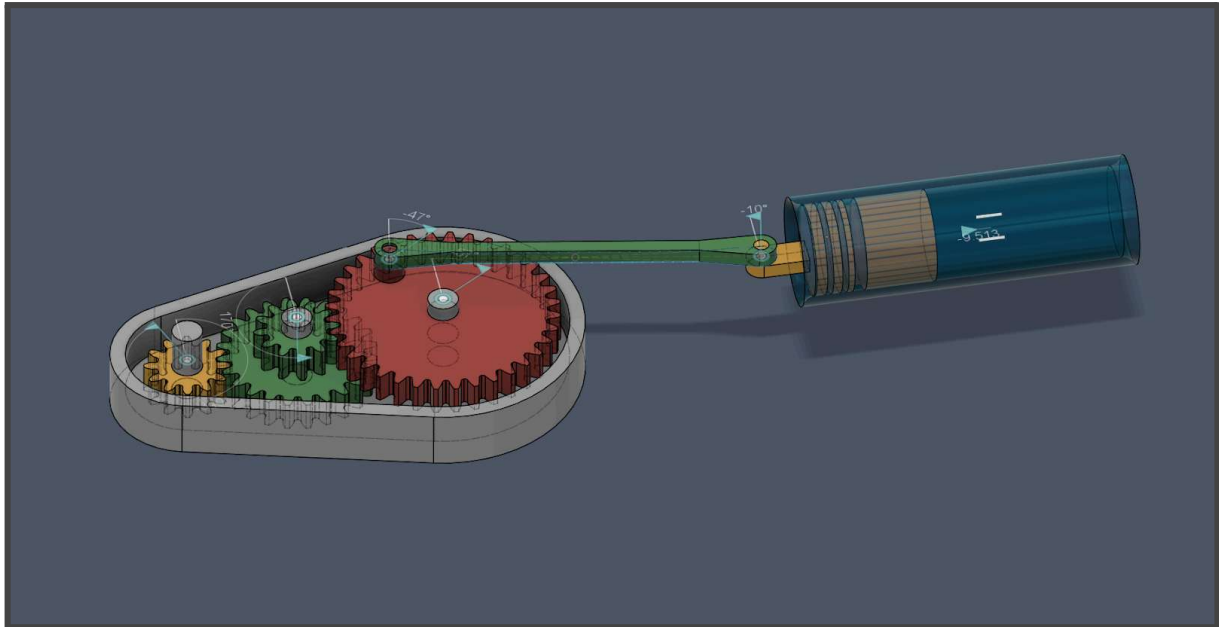
6th lab: Working with gears and motion studies

0. Resources

1. [How to Design Gears for 3D Printing](#) - How to use SpurGear (3 min)
2. [Motion Study Basics](#) - How to use Motion Studies (8 min)
3. [Motion Link Tutorial](#) - How to use Motion Links (9 min)
4. [Gear and Slider Modeling](#) - Project similar to this lab (19 min)

Today's design:

This is what we will be creating today:



Video:  Lab 6.mp4

1. Creating gears

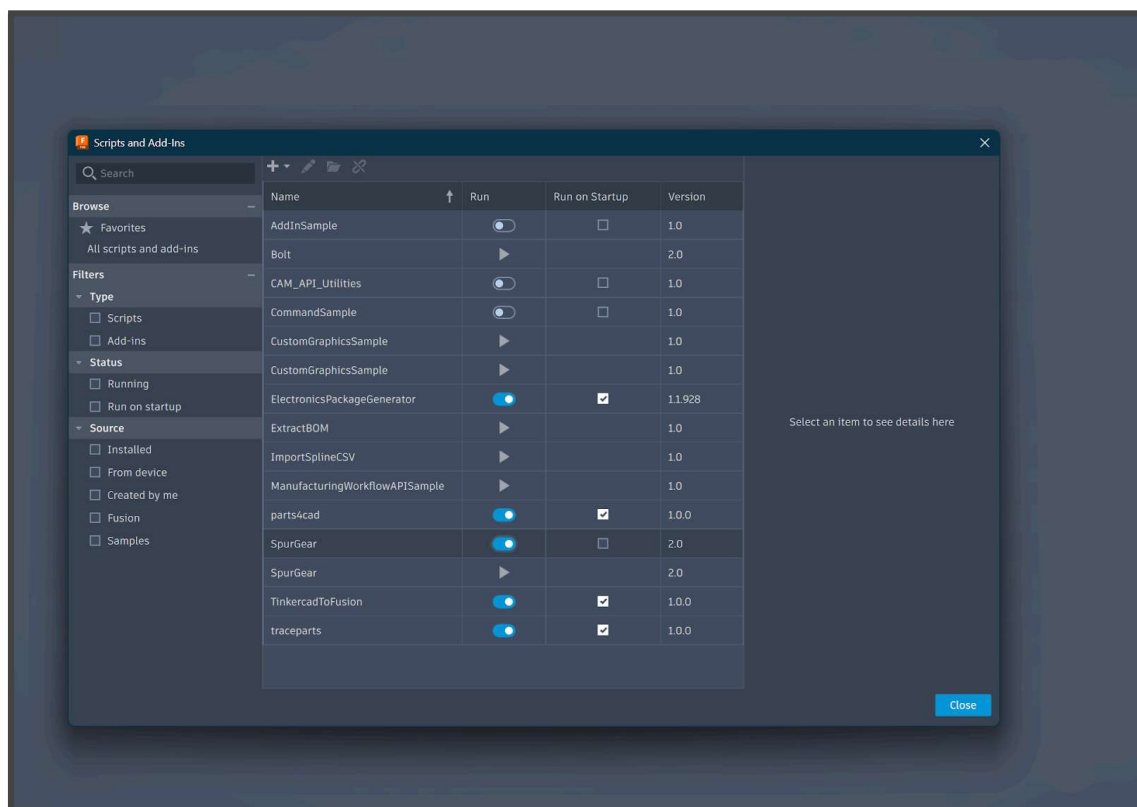
In order to create our first gear mechanism we will need to obtain some spur gears. The obvious solution would be to create a new sketch, design the gear and extrude it as we always did. However, as you are about to learn in the next exercises, the geometry of a basic gear is quite complex and we would be wasting too much time trying to replicate it.

Fusion 360 does not have a fully native gear creation tool, but does come with a set of pre-installed **Add-Ins**.

1.1 Spur Gear Add-In

Fusion 360 plugins, also called add-ins or scripts, are extensions (python scripts) created by the community to add specialized tools and commands. You can access the pre-installed plugins or install new ones via **Utilities > Add-Ins** in the tool bar. If you wish to follow along, everything will be done in **Hybrid design**, but you can also use **Assembly + Part design**.

You can also access the **Add-Ins** menu with the keyboard shortcut Shift + S.



The **Add-Ins** menu should look similar to the image above. Today we will be using **SpurGear**, which you can toggle on from here.

You can run the script from the button or go to **Create > Spur Gear**.

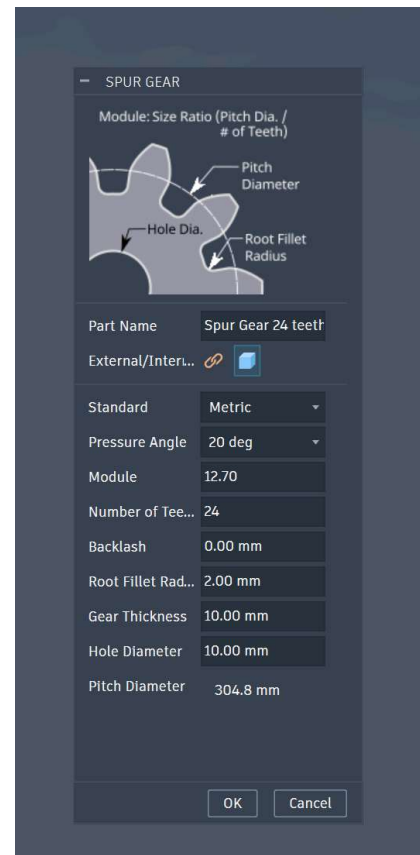
1.2 What is a spur gear? What defines it?

A spur gear is the simplest type of cylindrical gear, featuring straight teeth cut parallel to the axis of rotation.

Our script will create a new component with a single body that is our gear. Before that we first need to adjust the settings to our liking. One of the first things you might spot in these settings is the lack of a “size” adjustment. The size of our gear will be calculated based on our other settings, like the number of teeth and their dimension.

Here’s a breakdown of the main settings:

- **Standard** - whether the given dimensions are in metric or imperial.
- **Number of teeth** - what the name says.
- **Module** - a multiplier that adjusts the gear tooth size. The larger the number, the stronger and bigger the tooth. **For two gears to be able to fit, the module must be exactly the same for each.**
- **Pressure Angle** - adjusts the shape of each tooth.
- **Backlash** - a designed clearance or tolerance between meshing gear teeth. For 3D printing the value should be around 0.10mm - 0.15mm to compensate for printer error.
- **Root Fillet Radius** - specifies how rounded the inside corners of the gear tooth should be. Higher values will make the gears smoother and stronger, but values too high will produce an error.
- **Gear Thickness** - specifies how tall the gear should be. The plugin will create a sketch of the gear and extrude it up to this height.
- **Hole Diameter** - extrudes a hole at the center of the gear. You can leave it on 0 if you don’t want a hole or wish to add your own.
- **Pitch Diameter** - is the **calculated** size of the gear. As you can see from the picture, the pitch circle is **NOT** the total size of the gear. Rather it defines the points along each tooth where the gear is supposed to be in contact with another. Two gears positioned correctly should have their respective pitch circles tangent to each other.

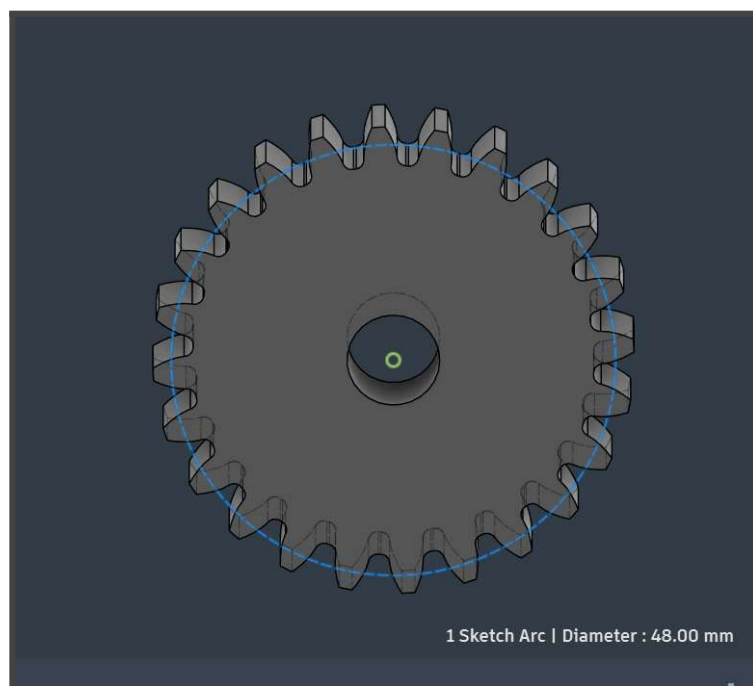
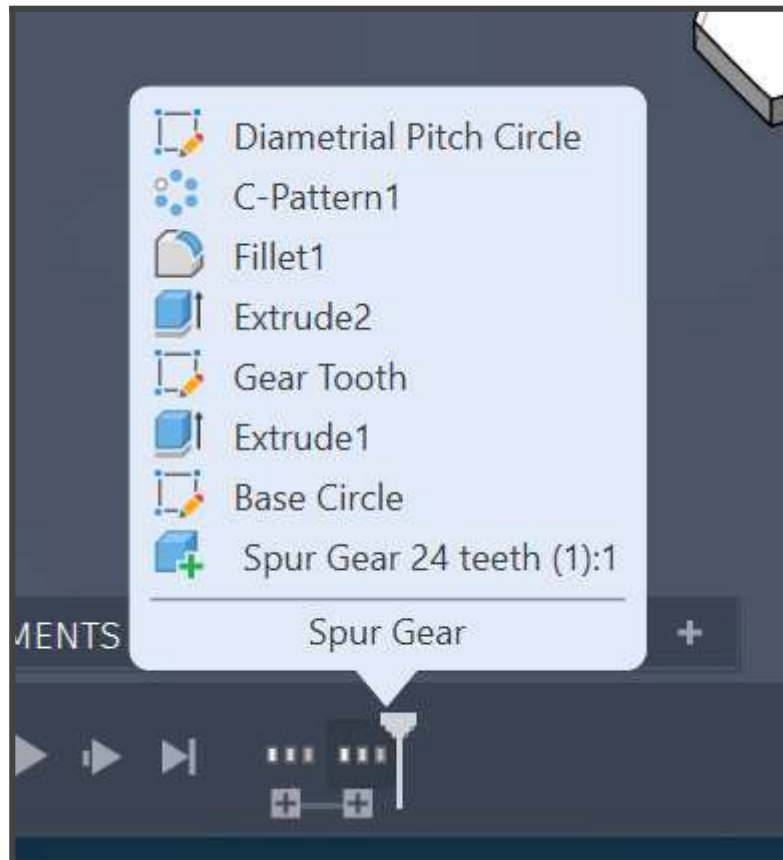


Based on this observation we can deduce that the correct distance between two gears is:

$$(\text{PD1} + \text{PD2}) / 2,$$

where **PD1** = Pitch Diameter of the first circle, **PD2** = Pitch Diameter of the second circle.

After creating the gear, you will see a new group of actions in the timeline. Each gear is created using operations we are already familiar with. You can expand this group by pressing the + symbol if you wish to see all of them in detail. The image below is what a gear will look like.

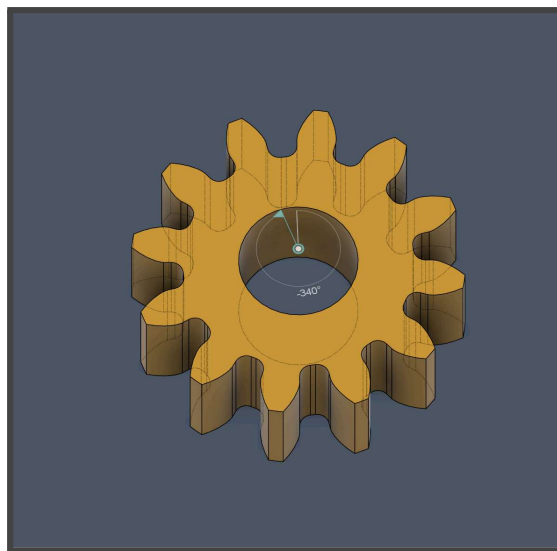


1.3 The components required:

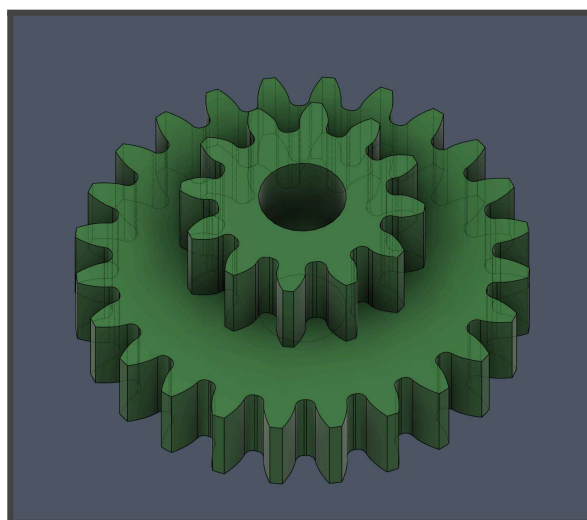
As before, exact dimensions of each component are not important. Just make sure they fit.

Remember, for two gears to work together, they need to have the same Module and Pressure Angle.

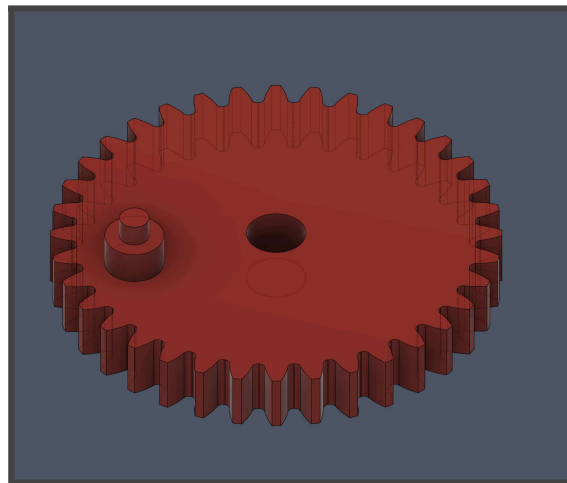
I. 12 tooth gear:



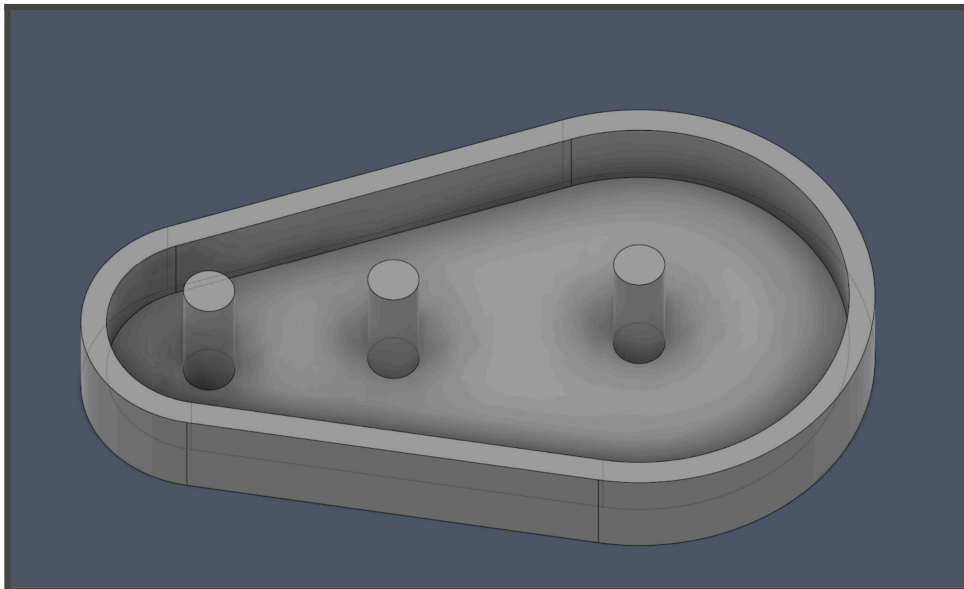
II. Double gear, a 12 tooth on top of a 24 tooth gear:



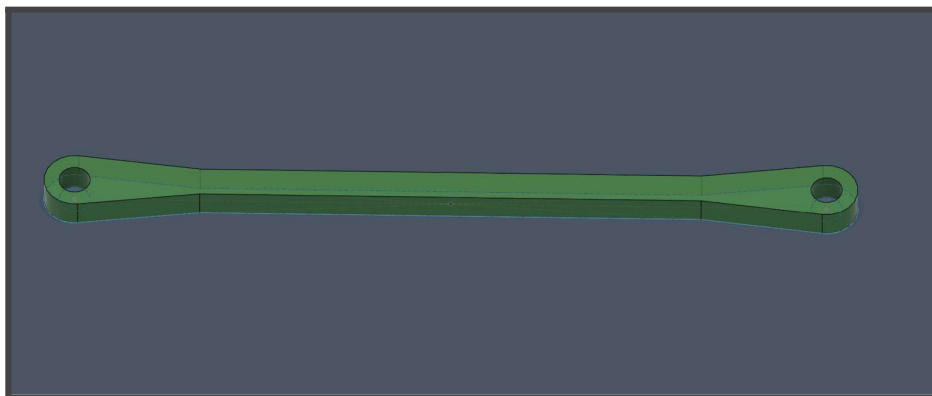
- III. 36 tooth drive gear, with a slot for a rod to attach:



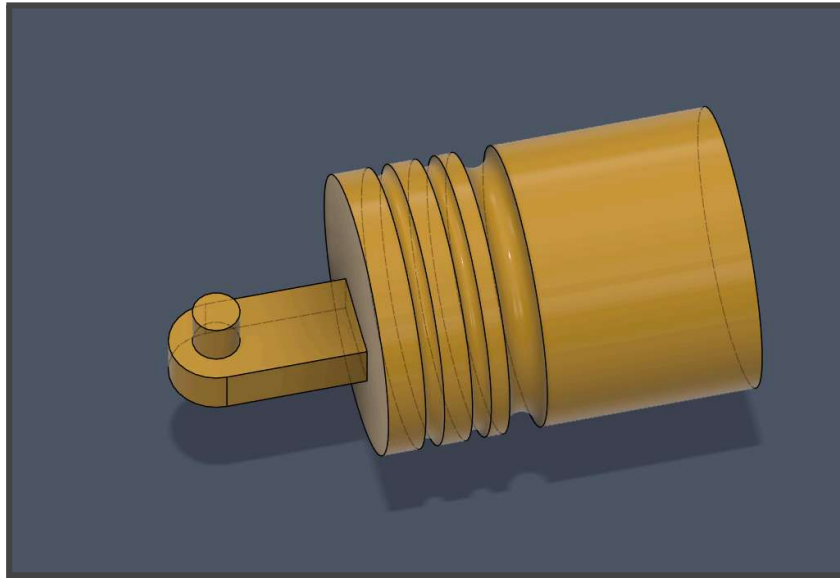
- IV. Case to house all gears. Remember the required distance between gears:



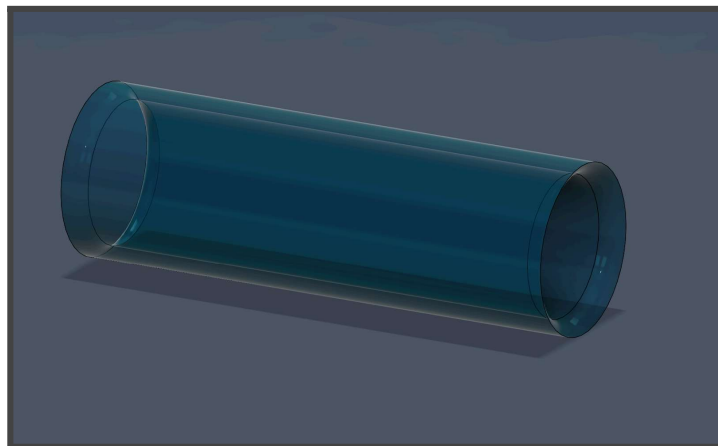
- V. Connection rod:



VI. Piston:

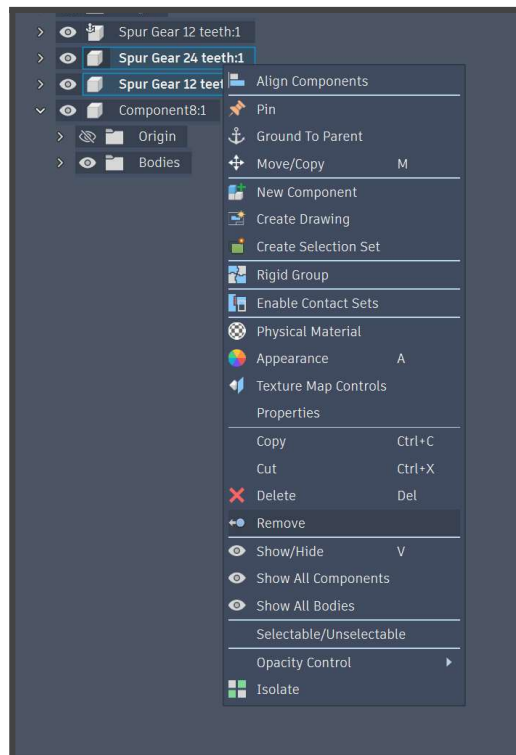


VII. Cylinder for piston to slide into:



1.4 Hints:

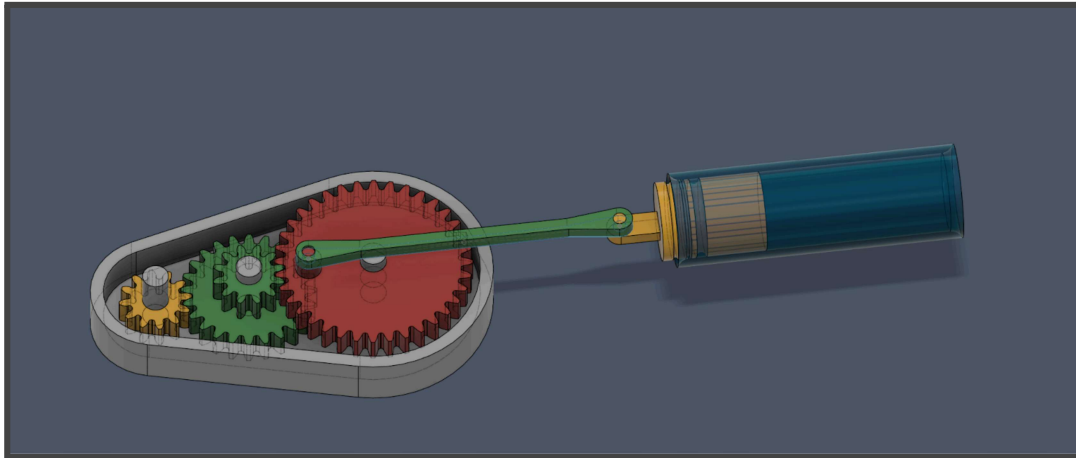
- Exact dimensions don't matter, as long as you make sure the parts fit.
- Each part should be a separate component.
- You can create the second gear by combining 2 components. Doing this from **Modify** > **Combine** will create a 3rd component. You can erase the now empty components without deleting them from the timeline by right clicking on them and selecting **"Remove"**:



- Keep track of the dimensions so the design makes sense.
- You can create additional sketches on the gears made using the Add-In.
- Remember to pin the case and cylinder to stop unwanted movement.
- You should use **Move** (M) type **"Point to Point"** to move the components into position.
- To get the gears into position you will need to rotate them enough so that their teeth fit between each other. For a 24 tooth gear, half a gear of rotation would be equal to $360 / 24 / 2$ degrees.

1.5 Joints:

After creating all components and moving them into position, your design should look something like this:



The next step is creating the joints between components. The 3 gears should revolve around the case, piston slider inside the cylinder, while the rod is supported by 2 revolute joints.

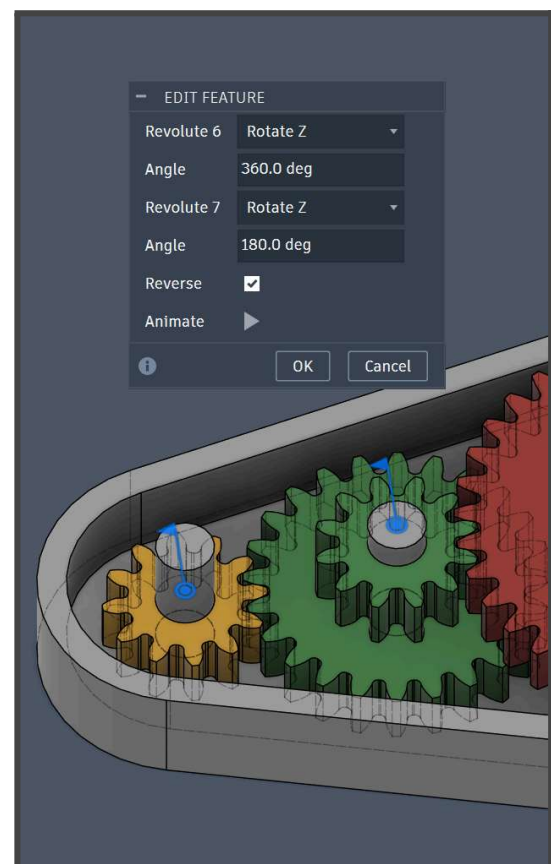
However, we can see that rotating one gear does not yet create the desired effect. In the complete design, rotating the yellow gear would move the piston. To solve that, we would need to add a **Motion Link** between the gears.

Go to **Assemble > Motion Link** to create one. You will be asked to select the two joints you want to link. For us, it's the revolute joints between the gears and the case. We will then need to calculate the relation between the movements.

In our example, as the yellow gear has 12 teeth and the green 24, one rotation of the first one would only correspond to half of a rotation on the second.

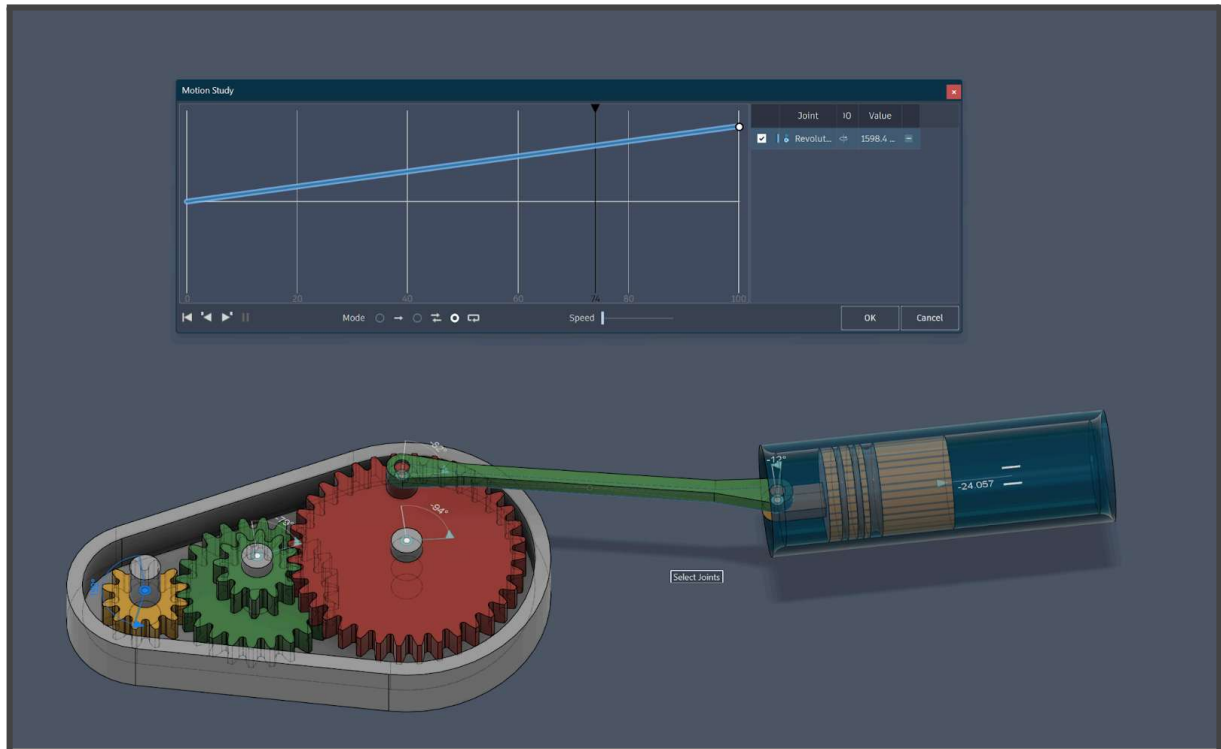
Therefore, for each 360 degrees of rotation at the first joint, Fusion should rotate the second 180 degrees, in the reverse direction.

A similar link is applied between the green (12 tooth) and red (36 tooth) gears.



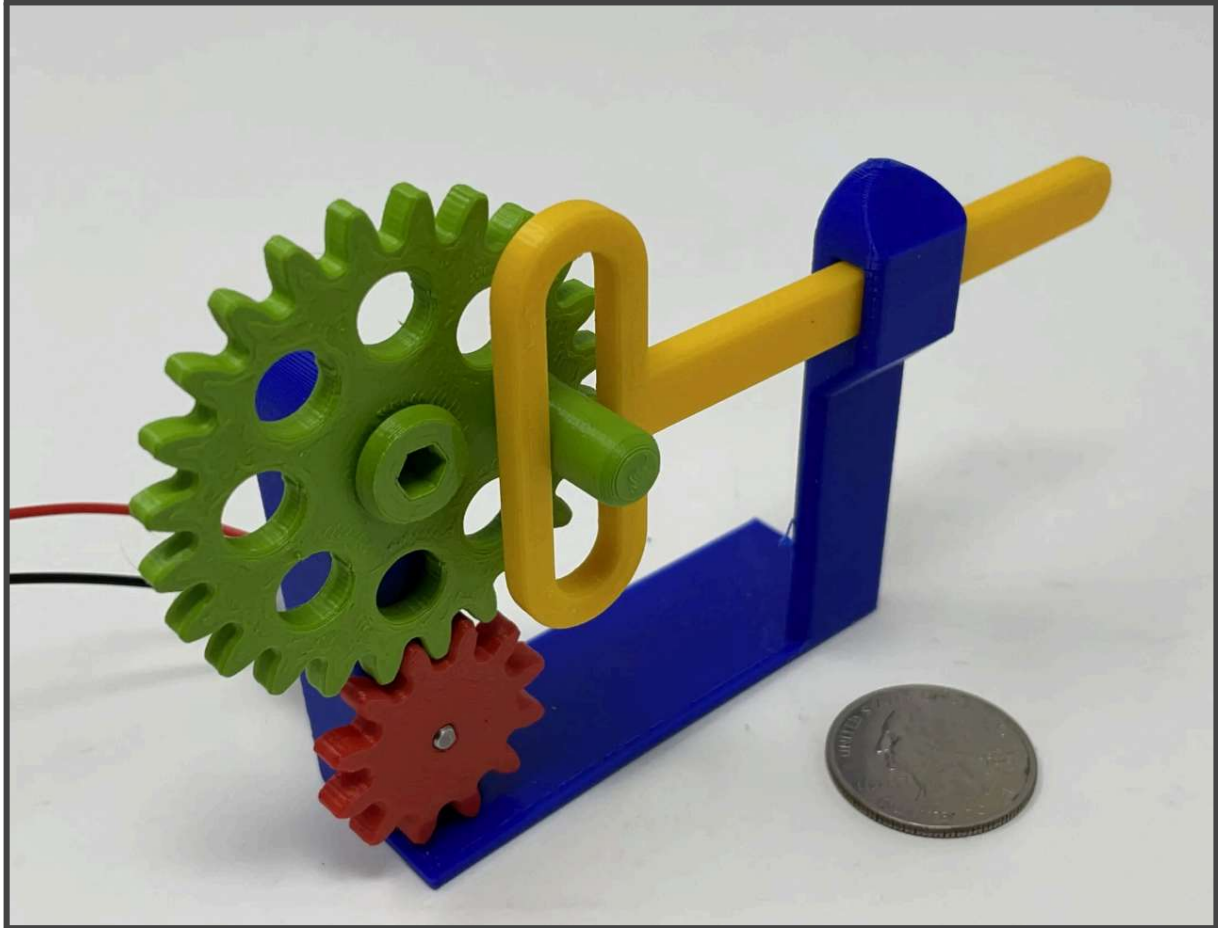
1.6 Motion study:

The design should now be complete. All that is left is creating a motion study that animates the yellow gear to activate the entire design:



2. Extra exercise

If you are finished with the previous design in time, you can also attempt to recreate the one below for a bonus:



[Simple 3D Printed Scotch Yoke](#)